

FORM TP 2018292



TEST CODE 02138020

MAY/JUNE 2018

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION*

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. ALL WORKING MUST BE SHOWN in this booklet.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of figures in answers will be penalized.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. Remember to draw a line through your original answer.
7. If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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SECTION A

Answer ALL questions.

1. (a) State Newton's first law of motion.

A body at rest remains at rest or if moving @ constant velocity in a straight line will continue to do so unless acted upon by an external (resultant) force.

[2 marks]

- (b) A student throws a ball of mass 0.02 kg vertically upwards into the air with an initial velocity of 25 m s^{-1} and catches it 6.0 seconds after the time of release. Table 1 shows how the velocity, v , of the ball changes with time, t , from the time of release to the time it is caught by the student.

TABLE 1: VELOCITY WITH TIME

Velocity, $v \text{ (m s}^{-1}\text{)}$	Time, $t \text{ (s)}$
25.0	0.0
15.0	1.0
6.5	2.0
0.5	3.0
-4.5	4.0
-14.0	5.0
-31.5	6.0

- (i) On the grid provided in Figure 1 on page 7, plot a graph of velocity, v , versus time, t . Draw the line of best fit through the points. [5 marks]

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V-axis: $\text{cm} \cdot \text{s}^{-1}$

Graph of v vs t .

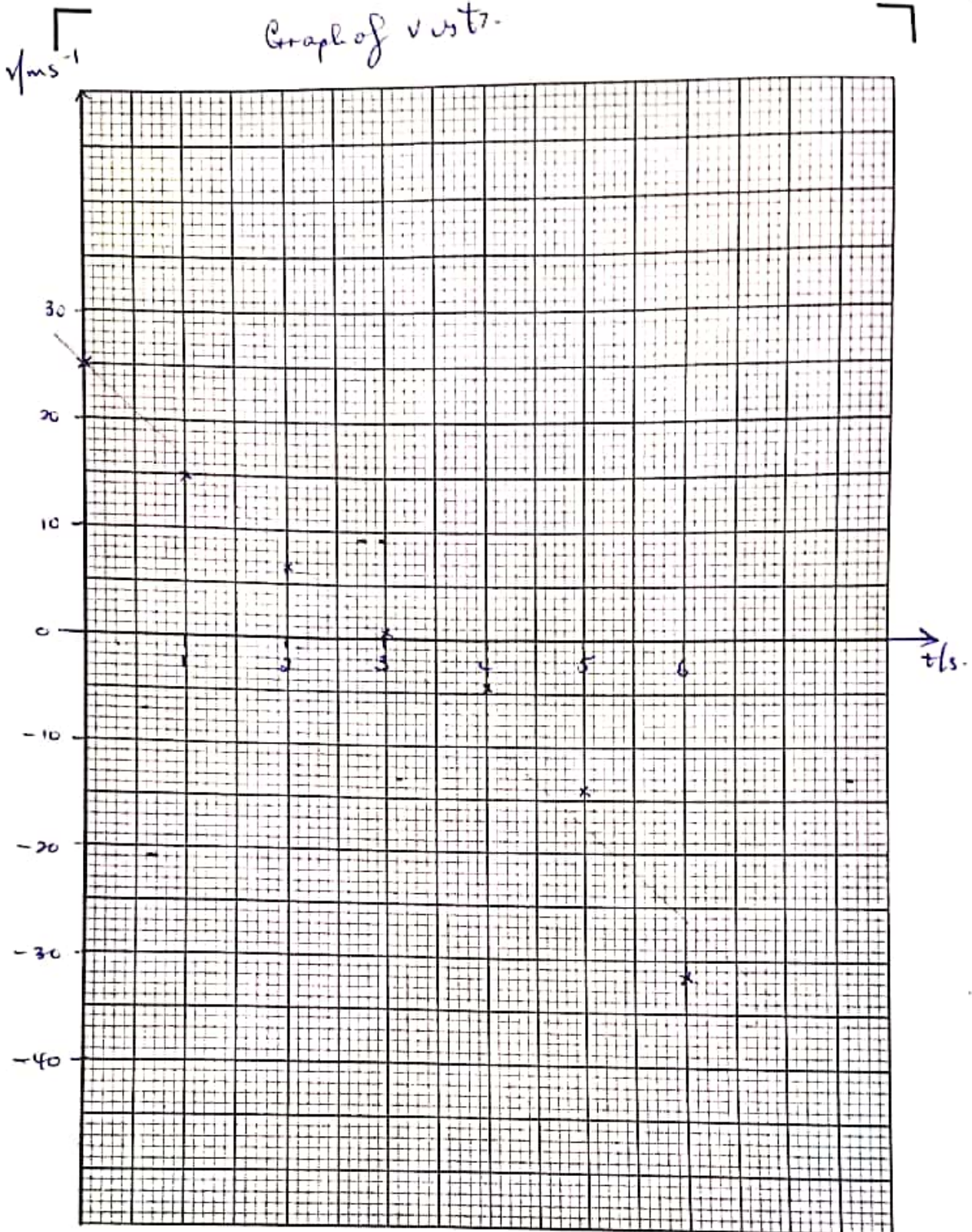


Figure 1. Velocity, v , versus time, t .

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- (v) The maximum potential energy calculated in (b) (iv) assumes that air resistance is negligible. Explain how this result would be affected if air resistance was significant.

The maximum potential energy would be less since the ball will lose energy as it rises due to air resistance. $\therefore$ h would be less.

[2 marks]

Total 15 marks

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- (ii) From your graph, determine the time at which the ball reaches its maximum height.

3 s.

[1 mark]

- (iii) Hence, or otherwise, determine the maximum height reached above the point of release.

Max height reached when ball is at rest
ie. $v = 0$.

Using $v^2 = u^2 + 2ax$
 $0^2 = 25^2 + 2 \times 9.81x$

$$x = \frac{-25^2}{2 \times 9.81}$$

$$x = 31.9 \text{ m}$$

From a $v-t$ graph this relates to the area under the curve.

$$\begin{aligned} x &= \frac{1}{2} b \times h \\ &= \frac{1}{2} \times 3 \times 25 \\ &= 37.5 \text{ m} \end{aligned}$$

[3 marks]

- (iv) Calculate the maximum potential energy gained by the ball in this throw.

$$\begin{aligned} E_p &= mgh \\ &= (0.02)(9.81) \times 37.5 \\ &= 7.36 \text{ J} \end{aligned}$$

[2 marks]

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- (b) The tone heard when a person blows across the top of a bottle partially filled with liquid (as shown in Figure 2) is due to cavity resonance.

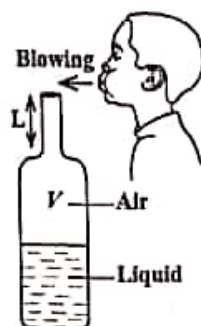


Figure 2. Cavity resonance

The frequency, f , of the tone depends on the volume, V , of air in the bottle and they are related by the following formula:

$$f = \frac{c}{2\pi} \sqrt{\frac{A}{LV}} \quad f = \frac{c}{2\pi} \sqrt{\frac{A}{L}} \cdot \frac{1}{\sqrt{V}}$$

Where c = velocity of sound in the air cavity in the bottle

A = cross-sectional area of the neck of the bottle

L = length of the neck of the bottle

V = volume of air in the air cavity in the bottle

In an experiment to investigate cavity resonance, the following values for f and V in Table 2 were obtained.

TABLE 2: FREQUENCY WITH VOLUME

Frequency f (Hz)	Volume V (m ³)	$\frac{1}{\sqrt{V}}$
225	250×10^{-6}	63.2
255	200×10^{-6}	70.7
290	150×10^{-6}	81.6
320	125×10^{-6}	89.4
350	100×10^{-6}	100
400	80×10^{-6}	112
445	65×10^{-6}	124

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2. (a) Define EACH of the following terms which are used to describe sound waves.

(i) Loudness

Loudness is a subjective quantity which relates to the amplitude of a sound produced (sound intensity).

(ii) Pitch

The pitch of a note is determined by its frequency.

(iii) Quality

The quality of a note is determined by the relative strengths of its overtones which varies from instrument to instrument.

[3 marks]

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(i) Complete Column 3 in Table 2 on page 12. [2 marks]

(ii) Using an appropriate scale, plot a graph of f against $\frac{1}{\sqrt{V}}$ on the grid provided in Figure 3 on page 13. Draw the line of best fit through the points. [4 marks]

(iii) Determine the gradient of the graph.

$$\begin{aligned} \text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \text{ using } (130, 455) \text{ and } (10, 35) \\ &= \frac{455 - 35}{130 - 10} \\ &= 3.50 \text{ s}^{-1} \text{ m}^{3/2} \end{aligned}$$

[3 marks]

(c) The bottle used in the experiment had a neck length of $5.8 \times 10^{-2} \text{ m}$. The cross-sectional area of the bottle's neck was $2.50 \times 10^{-4} \text{ m}^2$. Use this data along with your answer in (b) (iii) to determine the velocity of sound in the bottle.

$$\text{From } f = \frac{c}{2\pi} \sqrt{\frac{A}{l}} \cdot \frac{1}{\sqrt{V}}$$

$$\frac{c}{2\pi} \cdot \sqrt{\frac{A}{l}} = 3.50$$

$$\therefore c = \frac{3.50 \times 2\pi}{\sqrt{A/l}}$$

$$= \frac{3.50 \times 2\pi}{\sqrt{\frac{2.50 \times 10^{-4}}{5.8 \times 10^{-2}}}}$$

$$= \frac{3.50 \times 2\pi}{0.0657}$$

$$c = 334.7$$

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$$\therefore c = 335 \text{ m s}^{-1}$$



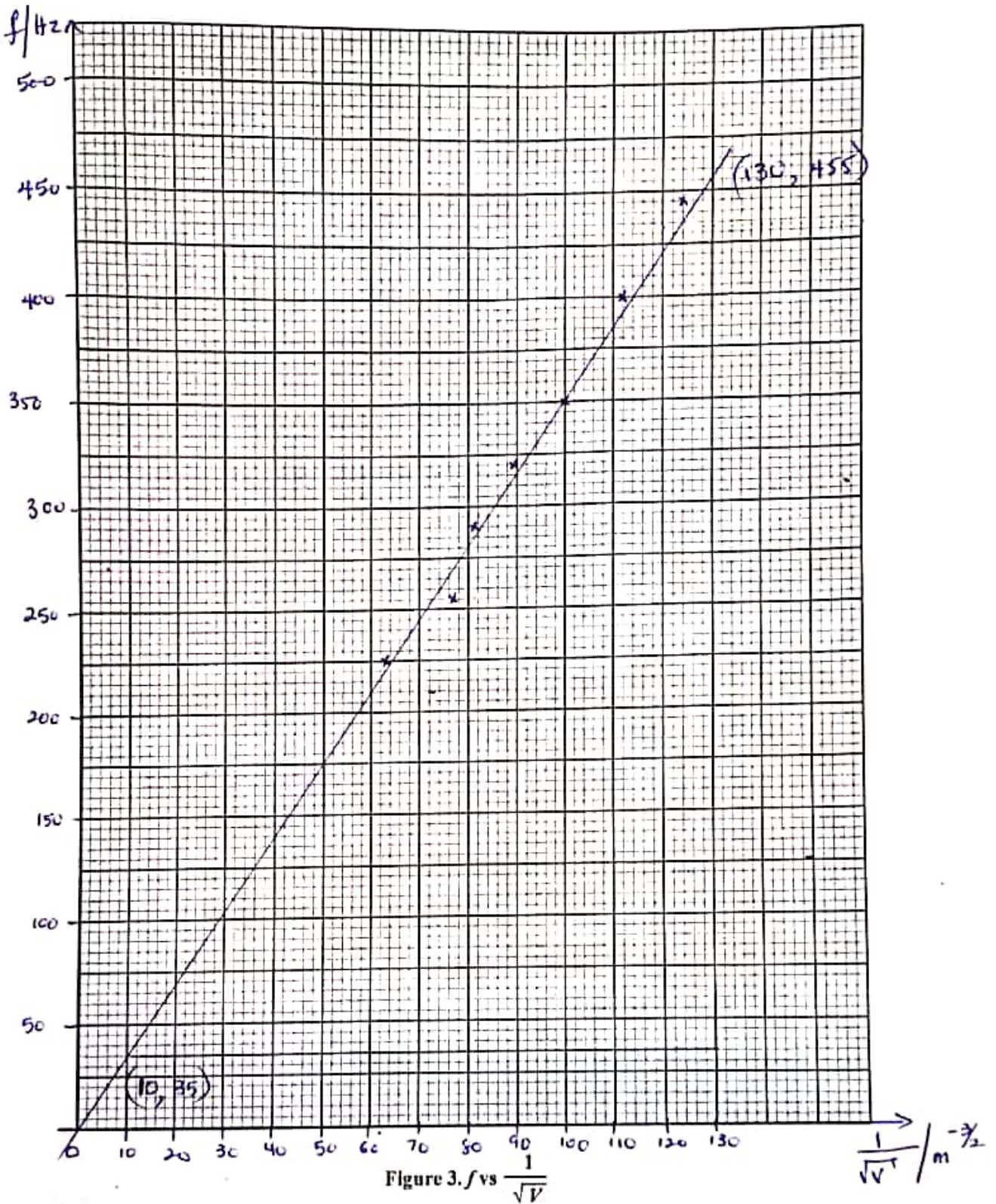
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[3 marks]

Total 15 marks

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Scale: f axis: 2cm = 50 Hz
 $\frac{1}{\sqrt{V}}$ axis: 1cm = 10 m^{-1/2}



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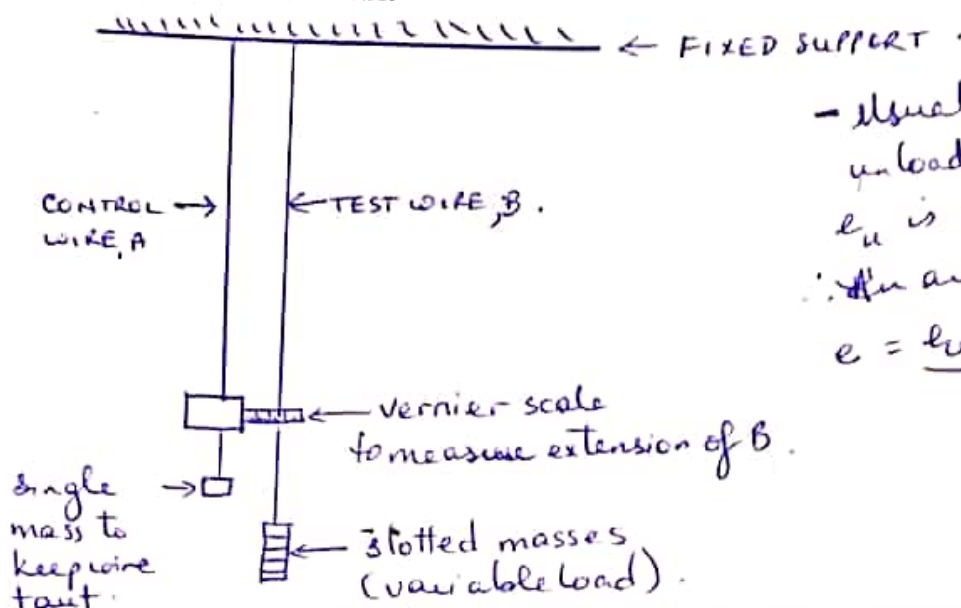
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3. (a) With the aid of a diagram, describe an experiment to accurately determine the Young's modulus of a metal wire.



- usually the wire is unloaded and an extension e_u is determined.
 ∴ the average extension $e = \frac{e_u + e_L}{2}$ is used

- The apparatus is set up as shown with both wires of the same material and the same length (2-4m). (P)
- The original length, l_0 , of the wire is measured from the fixed support to the beginning of the vernier scale.
- The diameter of the wire is measured spirally @ equidistant intervals along it, and the average diameter is found. (P)
- Weights are added, 10 N at a time, up to about 100 N and the extension, e_L , of the test wire is taken with reference to the vernier scale.
- Young's Modulus, $E = \frac{Fl_0}{Ae} = \left(\frac{F}{e}\right) \cdot \frac{l_0}{A} = k \cdot \frac{l_0}{A}$
 ∴ A graph of F vs e gives a gradient k

[6 marks]

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- (b) Figure 4 is a plot of the results from an experiment in which a metal wire was stretched.

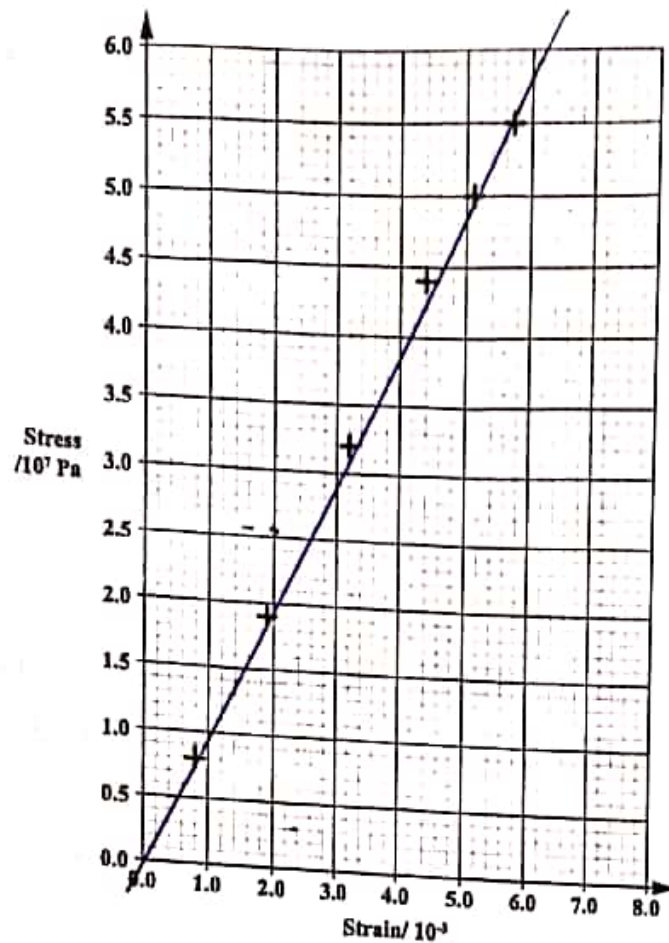


Figure 4. Stress vs strain for metal wire

- (i) Draw the line of best fit through the data points. [1 mark]
- (ii) If the area under the graph represents the energy stored per unit volume of the wire, determine the energy stored in one cubic millimetre of the wire.

$$\begin{aligned}
 \text{Area} &= \frac{1}{2} b \times h \\
 &= \frac{1}{2} 6.2 \times 10^{-3} \times 6 \times 10^7 \\
 &= 186,000 \text{ J m}^{-3} \\
 &= 1.86 \times 10^{-4} \text{ J mm}^{-3}
 \end{aligned}$$

[3 marks]



SECTION B

Answer ALL questions.

4. (a) Discuss the nature, cause and effect of frictional forces.

Nature: Frictional forces oppose the relative motion of two bodies. These forces act parallel to the surfaces in contact. Subdivided into static + dynamic friction.

Cause: Weight of body and surface roughness of faces in contact. Greater roughness, larger the frictional force.

Effects: Heat energy is produced and it retards motion (body slows down).

[6 marks]



- (c) A cylindrical copper wire and a cylindrical steel wire, each of length 1.5 m and diameter 2 mm, are joined to form a composite wire 3 m long. The composite wire is loaded until its length becomes 3.0003 m. $e = 0.0003 \text{ m}$

Young's modulus for copper = $1.2 \times 10^{11} \text{ N m}^{-2}$

Young's modulus for steel = $2.0 \times 10^{11} \text{ N m}^{-2}$

- (i) Calculate the ratio of the strain in the steel wire to the strain in the copper wire.

$$E_c = \frac{\text{stress}_c}{\text{strain}_c} \quad \text{and} \quad E_s = \frac{\text{stress}_s}{\text{strain}_s}$$

Since the wire is composite, the stress is the same.

$$\therefore E_c \text{ strain}_c = E_s \text{ strain}_s$$

$$\begin{aligned} \therefore \frac{\text{strain}_s}{\text{strain}_c} &= \frac{E_c}{E_s} \\ &= \frac{1.2 \times 10^{11}}{2.0 \times 10^{11}} \\ &= 0.6 \end{aligned}$$

[2 marks]

- (ii) Hence, calculate the extension of the copper wire.

$$\frac{\text{strain}_s}{\text{strain}_c} = \frac{1.2}{2.0}$$

$$e_c = \frac{5}{8} (0.0003)$$

$$e_c = 1.875 \times 10^{-4} \text{ m}$$

[3 marks]

Total 15 marks

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- (d) A particle of mass 2.0 kg is moving with SHM of period 8.0 s and amplitude 5.0 m. For this particle, calculate

- (i) the maximum attainable speed of the particle

$$v = x_0 \omega, \quad \omega = 2\pi f, \quad f = \frac{1}{T} = \frac{1}{8}$$

$$v = 5(2\pi \times 0.125)$$

$$v = 3.93 \text{ ms}^{-1}$$

[1 mark]

- (ii) the maximum energy attained by the particle.

$$\begin{aligned} E_k &= \frac{1}{2} m v^2 \\ &= \frac{1}{2} (2) (3.93)^2 \\ &= 15.4 \text{ J} \end{aligned}$$

[2 marks]

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5. (a) Define the term 'simple harmonic motion' (SHM).

SHM is motion in which the acc. of a particle is always directed towards and is proportional to its distance, x , from a fixed point.
 $a \propto -x$, $a = -\omega^2 x$, where $\omega^2 = \text{const.}$

[2 marks]

- (b) Write the mathematical expression relating acceleration, a , to displacement, x , in simple harmonic motion.

$$a = -\omega^2 x$$

[2 marks]

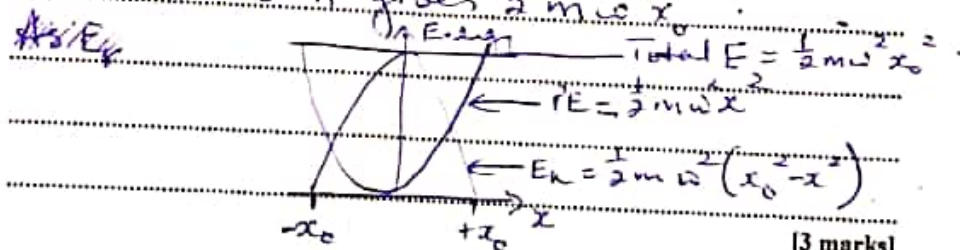
- (c) Describe the interchange of the kinetic energy and potential energy of an oscillating system during simple harmonic motion.

Total mechanical energy = $E_k + E_p = \frac{1}{2} m \omega^2 x_0^2 = \text{const.}$
 The principle of conserv. of energy holds since the total energy is const.

In SHM: $P.E = \frac{1}{2} m \omega^2 x_0^2 \sin^2 \omega t$

$K.E = \frac{1}{2} m \omega^2 x_0^2 \cos^2 \omega t$

When added it gives $\frac{1}{2} m \omega^2 x_0^2$.

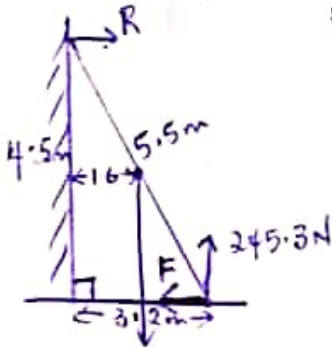


[3 marks]



- (b) A uniform ladder 5.5 m in length with a mass of 25 kg leans with its upper end against a smooth vertical wall. Its lower end rests on the rough ground. The ladder is in equilibrium when its bottom end is 3.2 m away from the wall and its top end rests against the wall.

- (i) Sketch a diagram to represent the arrangement described in the scenario above showing ALL the forces acting and ALL the distances described.



$$W = 25 \times 9.81$$

$$= 245.25 \text{ N}$$

[4 marks]

- (ii) Calculate the frictional force between the ladder and the ground.

Ladder is in equilibrium $\therefore$ resultant torque about any

$$(F \times 4.5) + 245.25 \times 1.6 = 245.3 \times 3.2$$

$$4.5F = 784.96 - 392.4$$

$$F = 87.2 \text{ N}$$

[5 marks]

Total 15 marks

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- (c) A mass of 0.5 kg is attached to a string of length 2.0 m. It is whirled in a horizontal circle at a speed of 5 m s⁻¹.

- (i) Calculate the period of oscillation for the mass.

$$T = \frac{2\pi}{\omega}, \quad C = \frac{mv^2}{r} = \frac{0.5 \times 5^2}{2} = 6.25 \text{ N}$$

$$= 2\pi \times 2.5$$

$$T = 2.5 \text{ s.}$$

$$6.25 = m\omega^2 r$$

$$\omega^2 = \frac{6.25}{0.5 \times 2}$$

$$\omega = 2.5 \text{ rad s}^{-1}$$

[3 marks]

- (ii) The mass is now suspended as a simple pendulum and made to oscillate. Determine the period of the resulting SHM.

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$= 2\pi \sqrt{\frac{2}{9.81}}$$

$$= 2.83 \text{ s.}$$

[2 marks]

Total 15 marks

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✓ $W = +$, work done to compress gas.

- (b) The gas is then compressed adiabatically to $3.4 \times 10^{-5} \text{ m}^3$ and its temperature rises to 795 K. Explain this rise in temperature

- (i) on a macroscopic scale, using the first law of thermodynamics

1st law: $\Delta U = Q + W$, $Q = 0$
 $\therefore \Delta U = +W$

$$\Delta U = \Delta E_k \propto \Delta T$$

$$\therefore +\Delta U \propto +\Delta T$$

Gas compressed when $Q = 0$, $\therefore$ work done on gas
 and the internal energy rises. $\therefore \Delta T \uparrow$.

[3 marks]

- (ii) on a microscopic scale, using the kinetic theory of gases.

gas is compressed $\therefore$ molecules are
 under a greater pressure to equilibrate
 to the external pressure.

$\therefore$ molecules move faster $\therefore E_k \uparrow \therefore \Delta T \uparrow$

[3 marks]

- (c) Calculate the pressure of the gas after the compression in (b).

$$\Delta U = \Delta W$$

$$\frac{3}{2}RT = P\Delta V$$

$$P = \frac{\frac{3}{2}RT}{\Delta V}$$

$$= \frac{\frac{3}{2} \times 8.31 \times 795}{3.4 \times 10^{-5} - 2.8 \times 10^{-4}}$$

$$3.4 \times 10^{-5} - 2.8 \times 10^{-4}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{1.03 \times 10^5 \times 2.8 \times 10^{-4}}{305} = \frac{P_2 \times 3.4 \times 10^{-5}}{795}$$

$$P_2 = 2.21 \times 10^6 \text{ Pa}$$

[2 marks]

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0.000246



6. Figure 5 shows a quantity of gas in a cylinder with a piston in thermal equilibrium at a temperature of 305 K. The initial volume of gas in the container was $2.80 \times 10^{-4} \text{ m}^3$ and its pressure was $1.03 \times 10^5 \text{ Pa}$.

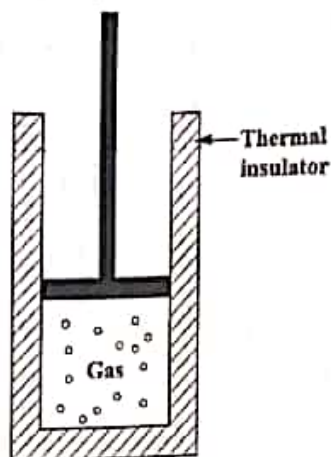


Figure 5. Gas in a cylinder

- (a) Use the equation of state for an ideal gas to determine the number of moles of gas in the cylinder.

$$\begin{aligned}
 PV &= nRT \\
 n &= \frac{PV}{RT} \\
 &= \frac{1.03 \times 10^5 \times 2.80 \times 10^{-4}}{8.31 \times 305} \\
 &= 0.0114 \text{ moles.}
 \end{aligned}$$

[3 marks]

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- (d) The work done on the gas during the compression is 85 J. Use the first law of thermodynamics to determine the INCREASE in the thermal energy of the gas during the compression.

$$\Delta U = \Delta W$$

$$\Delta U = 85 \text{ J}$$

[1 mark]

- (e) Rather than compressing the gas as in (b), the same mass of the same gas is heated in the same cylinder at constant volume, starting with the same initial conditions. Determine its molar heat capacity, C_v , if the heat supplied is 85 joules and the final temperature is 795 K.

$$E = n C_v \Delta \theta$$

$$C_v = \frac{E}{n \Delta \theta}$$

$$= \frac{(85)}{(0.0114)(795 - 305)}$$

$$= 15.2 \text{ J mol}^{-1} \text{ K}^{-1}$$

[3 marks]

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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